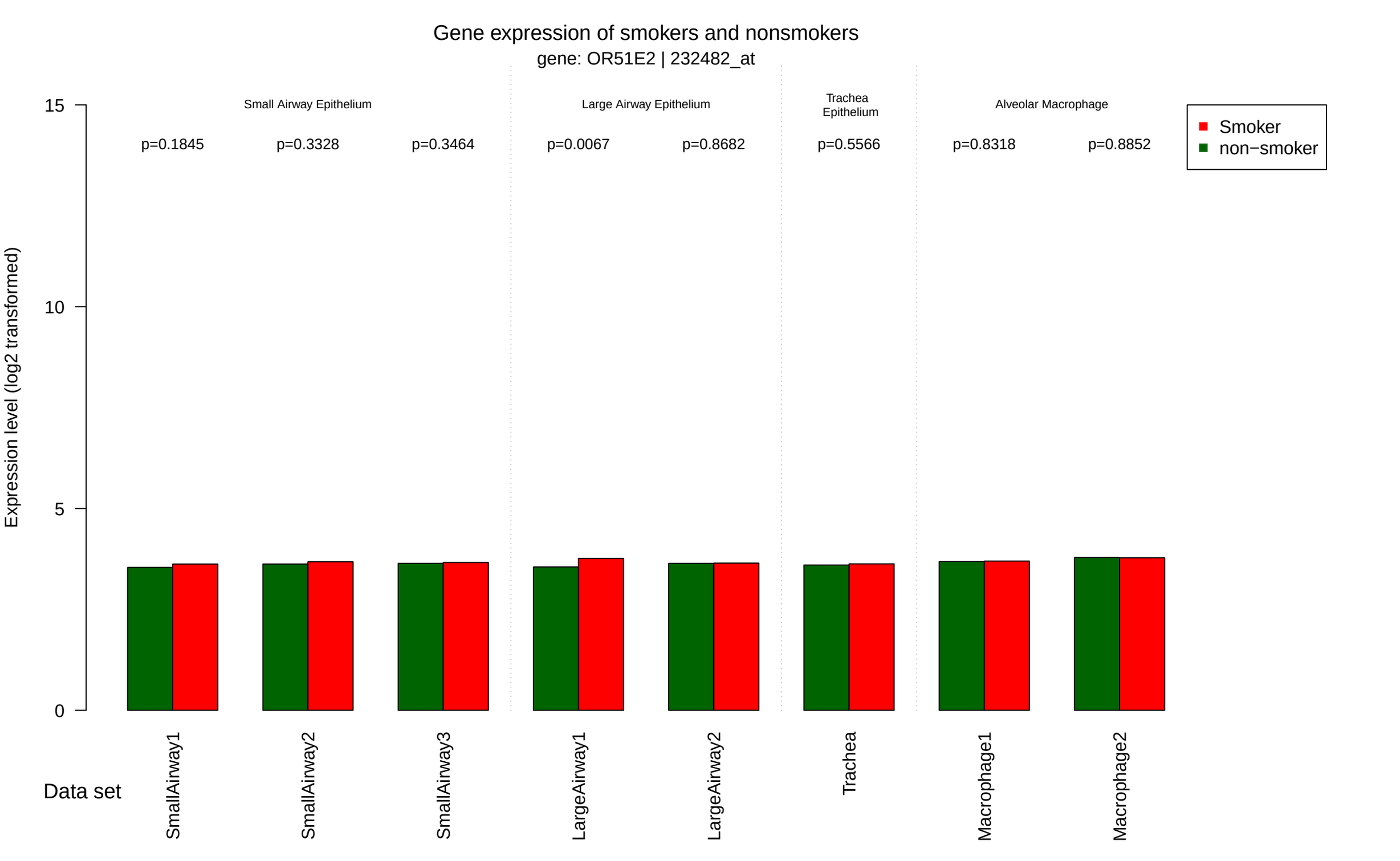
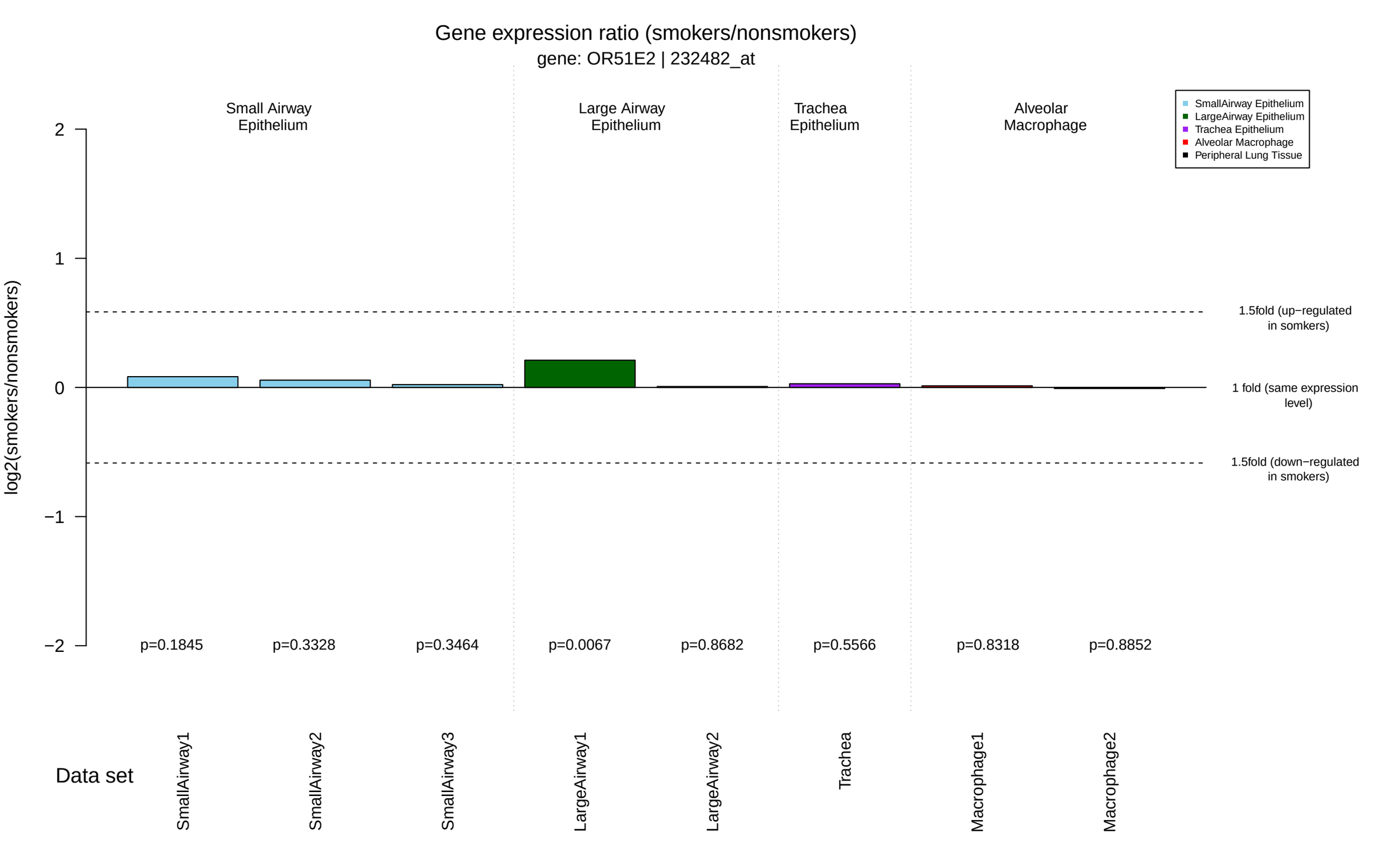
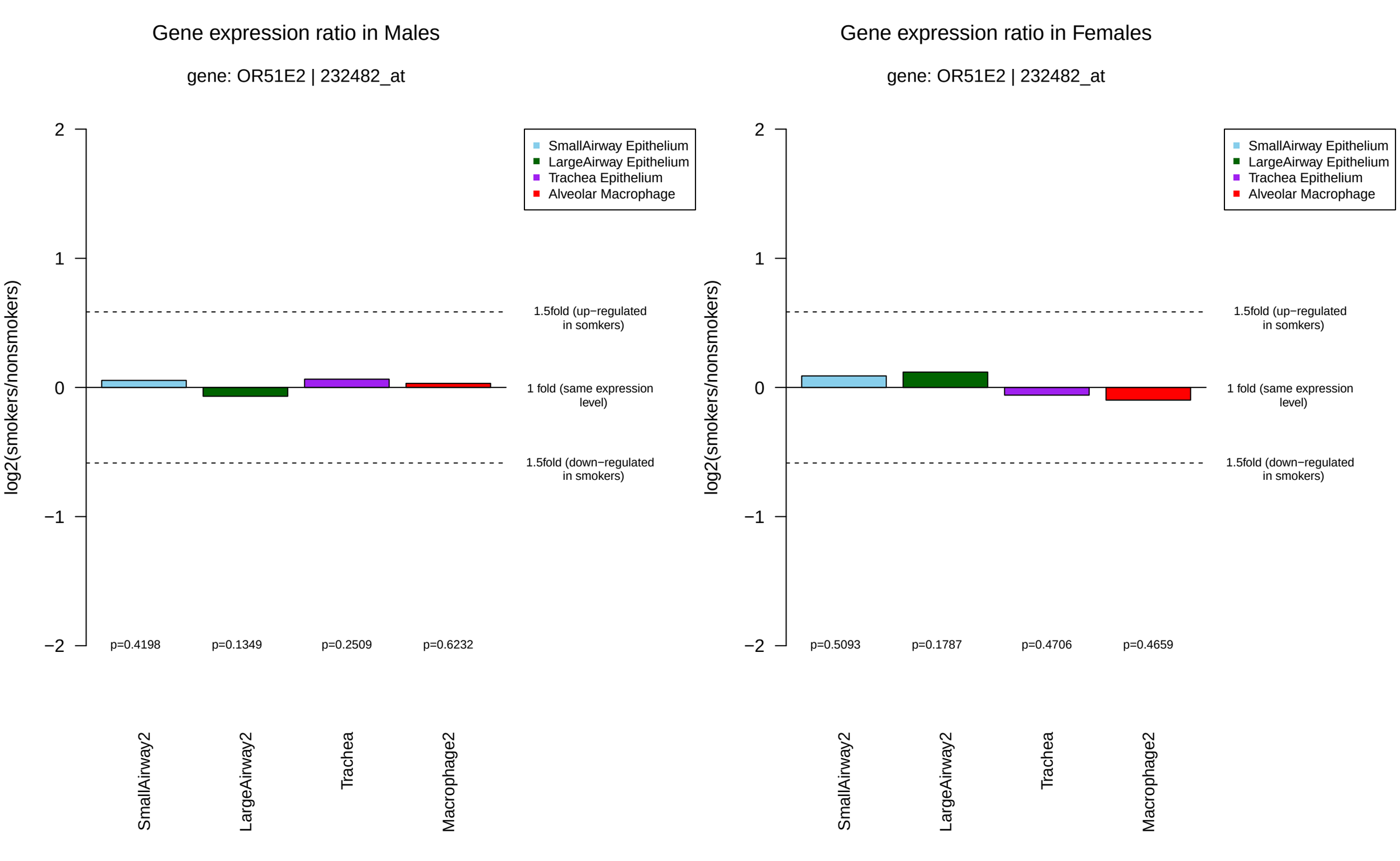
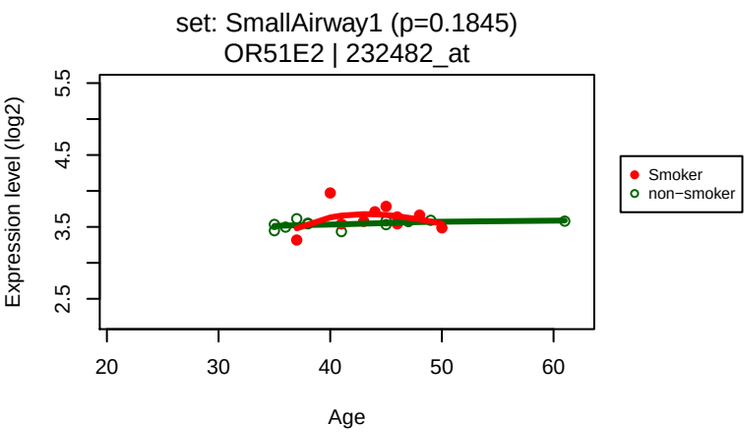
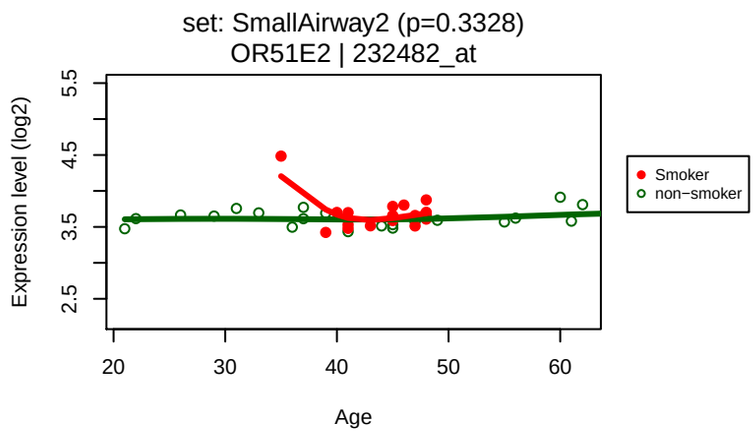
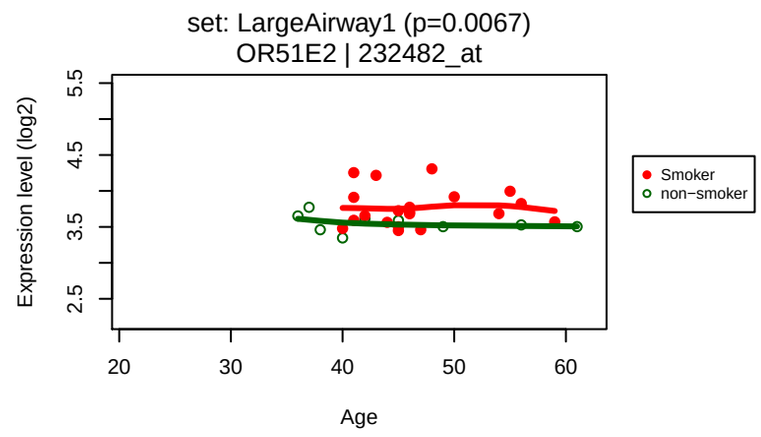

Gene expression of smokers and nonsmokers
gene: OR51E2 | 232482_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1845
p=0.3328
p=0.3464
p=0.0067
p=0.8682
p=0.5566
p=0.8318
p=0.8852
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

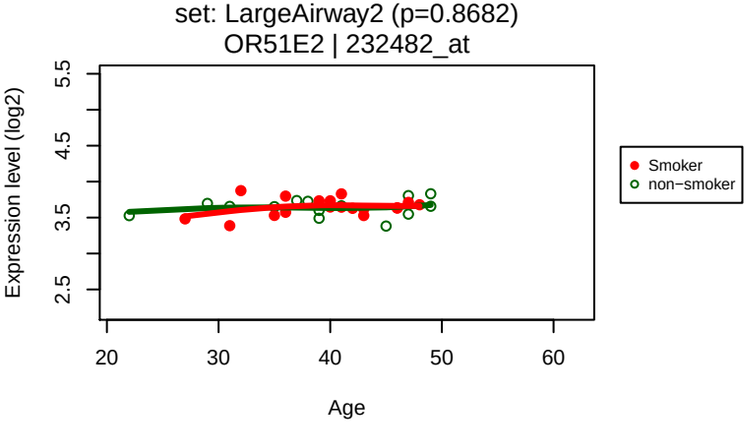

Gene expression ratio (smokers/nonsmokers)
gene: OR51E2 | 232482_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1845
p=0.3328
p=0.3464
p=0.0067
p=0.8682
p=0.5566
p=0.8318
p=0.8852
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

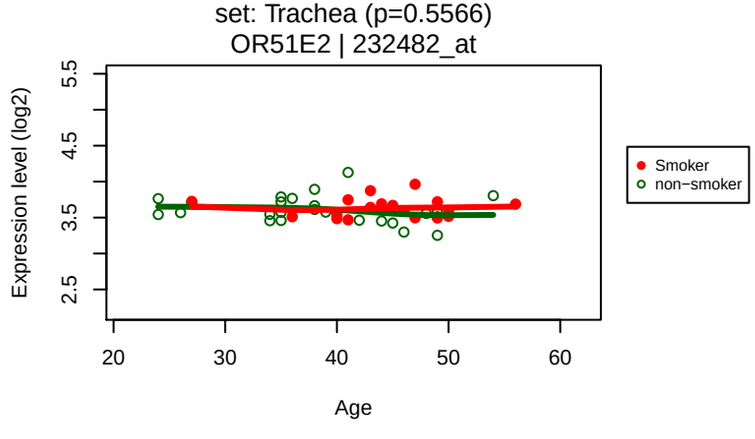

Gene expression ratio in Males
gene: OR51E2 | 232482_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4198
p=0.1349
p=0.2509
p=0.6232
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR51E2 | 232482_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5093
p=0.1787
p=0.4706
p=0.4659
SmallAirway2
LargeAirway2
Trachea
Macrophage2

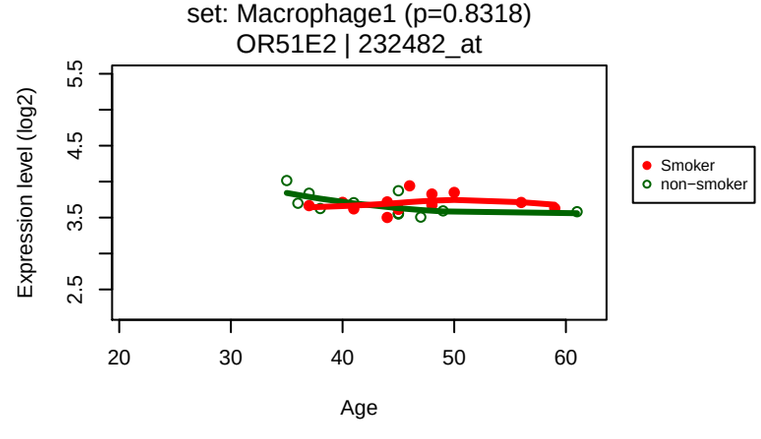

set: SmallAirway1 (p=0.1845)
OR51E2 | 232482_at
Expression level (log2)
Age
Smoker
non−smoker

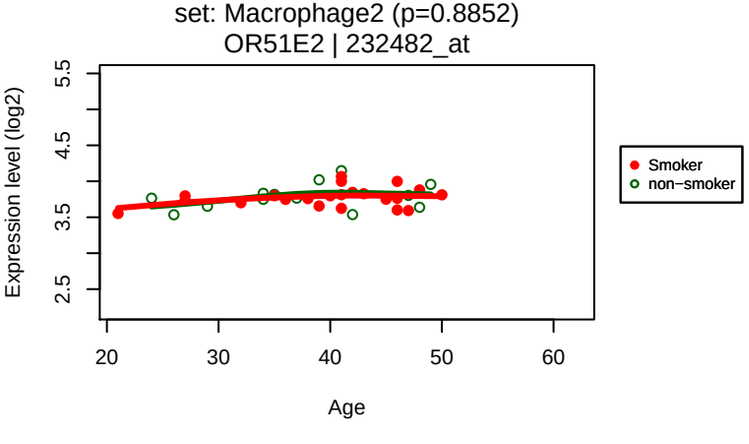

set: SmallAirway2 (p=0.3328)
OR51E2 | 232482_at
Expression level (log2)
Age
Smoker
non−smoker

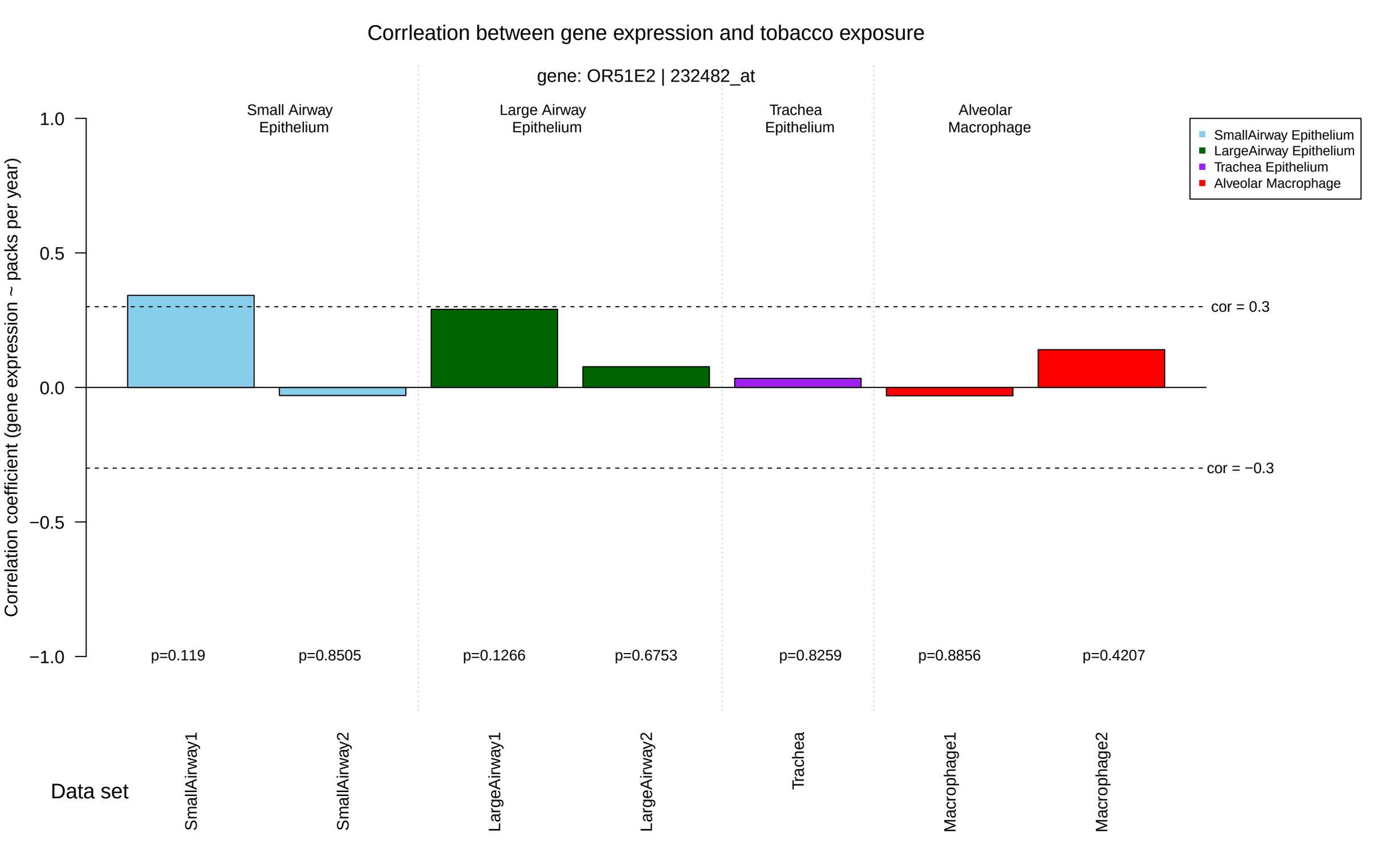

set: LargeAirway1 (p=0.0067)
OR51E2 | 232482_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8682)
OR51E2 | 232482_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5566)
OR51E2 | 232482_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8318)
OR51E2 | 232482_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8852)
OR51E2 | 232482_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR51E2 | 232482_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.119
p=0.8505
p=0.1266
p=0.6753
p=0.8259
p=0.8856
p=0.4207
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2